

Maps & Locations Reference for GS1

UPSC Mains General Studies Paper 1 — Geography,
Resources & Human Geography

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1. River Systems of India

1.1 Himalayan River Systems

Indus River System

Origin: Lake Mansarovar, Tibet | **Length:** 3,180 km (709 km in India) | **Mouth:** Arabian Sea near Karachi

Key Facts: Indus Water Treaty (1960) allocates eastern rivers (Ravi, Beas, Sutlej) to India and western rivers (Indus, Jhelum, Chenab) to Pakistan. Indus enters India through Ladakh. Tributaries: Zaskar (Ladakh), Jhelum (Vernag Spring, J&K), Chenab (Baralacha Pass, HP — longest tributary, 1,200 km), Ravi (Rohtang Pass), Beas (Beas Kund, HP), Sutlej (Lake Rakshastal, Tibet — enters via Shipki La).

Major Dams: Bhakra Nangal (Sutlej, 226m — 2nd highest in Asia), Pong (Beas), Salal (Chenab), Baglihar (Chenab).

Tributary	Origin	Length (km)	States
Jhelum	Vernag Spring (Kashmir)	725	J&K
Chenab	Baralacha Pass (Himachal)	1,200	HP, J&K
Ravi	Rohtang Pass (Himachal)	720	HP, Punjab
Beas	Beas Kund (Himachal)	470	HP, Punjab
Sutlej	Lake Rakshastal (Tibet)	1,450	HP, Punjab

Ganga River System

Origin: Gangotri Glacier (Gaumukh, Uttarakhand) as Bhagirathi | **Length:** 2,525 km | **Mouth:** Bay of Bengal (Sunderbans delta — world's largest)

Key Facts: Confluence of Bhagirathi + Alaknanda at Dev Prayag. Alaknanda (hydrologically longer) joins from Satopanth Glacier. Major tributaries: Yamuna (Yamunotri Glacier — 1,376 km, longest), Ghaghara (Tibet), Gandak (Nepal), Kosi ('Sorrow of Bihar'), Son (Amarkantak — only south bank), Ramganga,

Damodar ('Sorrow of Bengal'). Ganga basin covers 26% of India's area. Namami Gange Mission (2014, Rs 20,000 Cr).

Major Dams: Tehri (Bhagirathi, 260.5m — highest in India), Farakka (WB, Hooghly feeder). No large dams on main Ganga.

Brahmaputra River System

Origin: Chemayungdung Glacier, Tibet (as Tsangpo) | **Length:** 2,900 km (916 km in India) | **Mouth:** Bay of Bengal (as Jamuna in Bangladesh)

Key Facts: U-turn at Namcha Barwa (7,756m). Enters India at Arunachal (as Siang/Dihang). Joins Dibang + Lohit at Sadia (Assam). Tributaries: Subansiri (412 km — largest in India), Manas, Sankosh, Teesta (309 km, Sikkim & Bangladesh). Highest sediment load in India. Causes annual Assam floods. Majuli Island (world's largest river island).

Major Dams: Subansiri Lower (2,000 MW, under construction), Teesta III-V (Sikkim).

1.2 Peninsular River Systems

River	Origin	Length (km)	Flow	Mouth	Major Dams
Godavari	Trimbakeshwar (Nasik)	1,465	East	Bay of Bengal (Rajahmundry)	Jayakwadi, Sriram Sagar, Polavaram
Krishna	Mahabaleshwar (Satara)	1,400	East	Bay of Bengal (Hamsaladeevi)	Koyna, Srisailem, Nagarjunasagar, Almatti
Cauvery	Talakaveri (Brahmagiri, KA)	800	East	Bay of Bengal (Poompuhar)	KRS, Mettur, Kabini, Kallanai (oldest)
Mahanadi	Pharsiya (Raipur, CT)	885	East	Bay of Bengal (False Point)	Hirakud (4.8 km, longest in India)
Narmada	Amarkantak (Anuppur, MP)	1,312	West		

				Arabian Sea (Gulf of Khambhat)	Sardar Sarovar, Indira Sagar, Bargi
Tapti	Betul (Madhya Pradesh)	724	West	Arabian Sea (Surat)	Ukai, Kakrapar

East vs West Flowing Rivers: East-flowing rivers (Godavari, Krishna, Cauvery, Mahanadi) carry 90% of peninsular drainage and form deltas. West-flowing rivers (Narmada, Tapti) flow through rift valleys between Vindhya-Satpura ranges, form estuaries (not deltas), and are shorter.

2. Mountain Ranges & Peaks

2.1 The Himalayas — Three Parallel Ranges

- **Himadri (Greater Himalayas):** Avg height 6,100m. Peaks: Everest (8,848m, Nepal), K2 (8,611m, POK), Kanchenjunga (8,586m, Sikkim — India's highest), Nanda Devi (7,817m, Uttarakhand), Kamet (7,756m). Glaciers: Siachen (76 km — world's highest battleground), Gangotri (30 km), Yamunotri, Zemu, Bara Shigri.
- **Himachal (Lesser Himalayas):** Height 3,700-4,500m. Hill stations: Shimla, Mussoorie, Nainital, Darjeeling. Ranges: Pir Panjal, Dhauladhar, Nag Tibba.
- **Shivaliks (Outer Himalayas):** Height 600-1,500m. Unconsolidated sediments. Dun valleys (Dehradun, Patli Dun) between Shivaliks and Himachal.

Major Himalayan Passes

Pass	State	Elevation (m)	Connects	Importance
Karakoram	Ladakh	5,540	India to Xinjiang (China)	Silk Route; Siachen nearby
Zoji La	Ladakh	3,528	Srinagar to Leh	Lifeline of Ladakh; closed winter
Shipki La	Himachal	4,570	India to Tibet (Sutlej corridor)	Border trade; Sutlej enters India here
Rohtang	Himachal	3,978	Manali to Leh	Tourist; Atal Tunnel bypass now
Nathu La	Sikkim	4,310	India to Tibet (Lhasa)	Border trade reopened 2006; Doklam nearby
Sela	Arunachal	4,170	Bomdila to Tawang	Strategic Tawang sector

2.2 Other Major Ranges

- **Western Ghats (Sahyadris):** UNESCO World Heritage (2012). 1,600 km Gujarat to Kanyakumari. Avg height 900-1,100m. Highest: Anamudi (2,695m, Kerala). Passes: Palghat, Shenkottah, Thal Ghat. Orographic rainfall on west-facing slopes. Origin of many peninsular rivers. Biodiversity hotspot.
- **Eastern Ghats:** Discontinuous, eroded hills. 1,100 km Odisha to TN. Avg height 600m. Highest: Arma Konda (1,680m, AP). Cut by east-flowing rivers. Key sections: Nallamala, Seshachalam, Shevaroy, Palni (Cardamom Hills).
- **Vindhya Range:** Separates North and South India. MP, UP, Bihar. Height 300-650m. Narmada and Son originate here.
- **Satpura Range:** South of Narmada, parallel to Vindhya. MP, Maharashtra, Gujarat. Height 900-1,350m. Highest: Dhupgarh (1,350m, Pachmarhi). Rich in minerals.
- **Aravalli Range:** Oldest fold mountain. 800 km Delhi to Gujarat. Highly eroded. Highest: Guru Shikhar (1,722m, Mount Abu, Rajasthan). Acts as climatic barrier — prevents desert expansion eastward.

Map Drawing Strategy: Draw India's outline and mark: (1) Himalayas as thick arc from Jammu to Arunachal, (2) Aravalli line Delhi to Gujarat, (3) Vindhya east-west across Central India, (4) Satpura below Vindhya, (5) Western Ghats along west coast, (6) Eastern Ghats as broken hills along east coast. Label 3 peaks and 4 passes on Himalayas.

3. Climate Zones & Rainfall Distribution

Koppen Climate Classification in India

Code	Climate Type	Regions in India
Am	Tropical Monsoon	Western Ghats, parts of NE India
Aw	Tropical Savannah	Peninsular India (Maharashtra, Karnataka, TN, AP)
BSh	Semi-Arid (Steppe)	Rajasthan, Gujarat (Kutch, Saurashtra)
BWk	Arid (Desert)	Western Rajasthan (Thar Desert)
Cwg	Monsoon with Dry Winter	Gangetic plains (UP, Bihar, WB)
Dsb/ Dwb	Humid Continental	Himalayan region (J&K, HP, Uttarakhand)
ET	Polar/Tundra	High Himalayas (above 4,500m)

Rainfall Distribution

Highest: Mawsynram (11,872 mm/yr) & Cherrapunji (11,430 mm) — Meghalaya, world's wettest. Western Ghats west slopes — 250-400 cm. NE states — 200+ cm.

Moderate: Gangetic plains — 100-200 cm. Central India — 75-150 cm.

Low: Rajasthan desert — below 25 cm. Rain shadow of WG (Deccan plateau) — 50-100 cm. Ladakh — 10-20 cm (cold desert).

Variability: El Nino reduces monsoon by 5-10%. La Nina enhances. Positive IOD counters El Nino effect.

4. Mineral, Oil & Gas Locations

Mineral	Major Producing States	Key Mines/Locations
Iron Ore	Odisha, Jharkhand, Karnataka, Chhattisgarh, Goa	Bailadila (CT — 68% Fe highest grade), Keonjhar (OD), Bellary-Hospet (KA)
Coal	Jharkhand, Odisha, Chhattisgarh, WB, MP, Telangana	Jharia (largest), Raniganj (oldest), Talcher (largest in OD), Singareni (TG)
Bauxite	Odisha, Gujarat, Jharkhand, Maharashtra, CT	Panchpatmali (OD — largest deposit), Jamnagar (GJ), Palamau (JH)
Copper	Rajasthan, MP, Jharkhand	Khetri (RJ), Malanjkhand (MP — largest open-pit), Singhbhum (JH)
Gold	Karnataka (only producing mine)	Hutti (KA — only active mine), Kolar (closed 2001)

Oil & Natural Gas Basins

Basin	Location	Type
Mumbai High	Offshore Arabian Sea	Largest oil field in India
Bassein	Offshore Maharashtra	Largest gas field
KG Basin	Andhra Pradesh (offshore)	Gas (Reliance KG-D6)
Assam-Arakan	Assam, Arunachal, Nagaland	Oil (Digboi — oldest in Asia)
Cambay Basin	Gujarat (Ankleshwar, Kalol, Mehsana)	Oil & Gas
Rajasthan Basin	Barmer, Jaisalmer	Oil (Mangala — largest onshore)

5. Agricultural Regions

Crop	Major Producing States	Conditions	Key Facts
Rice	WB (largest), UP, Punjab, Odisha, AP, Bihar, Chhattisgarh, Assam	20-30 deg C, 150-300 cm rain, alluvial/clay soil	India: 2nd largest producer. Kharif (80%) + Rabi (20%). 'Granary of India' — Punjab.
Wheat	UP (largest), Punjab, Haryana, MP, Rajasthan, Bihar	10-25 deg C, 50-100 cm rain, loamy soil	2nd largest producer. Rabi crop. 'Wheat bowl of India' — Punjab.
Sugarcane	UP (40%+), Maharashtra, Karnataka, TN, Bihar	21-27 deg C, 75-150 cm, deep loamy soil	2nd largest producer. 12-month crop.
Cotton	Gujarat (largest), Maharashtra (largest area), Telangana, Karnataka, MP	21-30 deg C, 50-100 cm, black soil (regur)	'Cotton Triangle' — GJ-MH-TG. 2nd largest producer. Bt cotton 95%+.
Tea	Assam (55%), WB (Darjeeling), TN (Nilgiri), Kerala (Munnar)	15-30 deg C, 150-250 cm, acidic soil	2nd largest producer. CTC (Assam), Orthodox (Darjeeling GI tag).
Coffee	Karnataka (70%+, Chikmagalur, Coorg), Kerala, TN	15-28 deg C, 150-200 cm, shade-grown	Arabica (35%) + Robusta (65%). GI: Coorg Arabica.
Jute	West Bengal (main), Assam	Hot, humid, alluvial soil (Ganga-Brahmaputra delta)	Largest producer globally.

Agricultural Belts

- **Rice-Wheat Rotation Belt:** Indo-Gangetic plains — Punjab, Haryana, UP, Bihar — the core of Green Revolution.
- **Cotton Triangle:** Gujarat-Maharashtra-Telangana — black cotton soil (regur).
- **Spice Regions:** Kerala (pepper, cardamom), Karnataka (cardamom), Rajasthan (cumin, coriander), Chhattisgarh (chilli), Kashmir (saffron).
- **Fruit Belts:** Maharashtra (grapes, Alphonso mango), UP (Malihabad mangoes), J&K (apple — Srinagar), HP (apple).

6. Industrial Regions & Hubs

Region	States	Key Industries	Location Factors
Mumbai-Pune	Maharashtra	Auto, IT, textiles, pharma, finance	Port, finance capital, skilled labour
Bangalore-TN	Karnataka, TN	IT/ITeS, aerospace, biotech, auto	Skilled workforce, climate, R&D institutions
Delhi-NCR	Delhi, UP, Haryana	Auto (Gurugram), IT (Noida), manufacturing	Political capital, market, infrastructure
Ahmedabad-Vadodara-Surat	Gujarat	Textiles (MMF), petrochemicals, diamonds (Surat 90% world cutting)	Ports, oil/gas, entrepreneurial culture
Chennai-Hyderabad-Bangalore	TN, Telangana, KA	IT corridor, auto (Chennai — Detroit of Asia), pharma (Hyderabad — Bulk Drug Capital)	Skilled labour, SEZs, climate
Jamshedpur-Ranchi-Dhanbad	Jharkhand	Steel (Tata Steel oldest), coal, heavy engineering	Mineral belt (coal, iron ore)
Kolkata-Haldia-Asansol	West Bengal	Jute, tea, steel, manufacturing	Port, mineral belt, historical base

Automobile Hubs

Chennai (Detroit of Asia — Ford, Hyundai, Renault-Nissan, Ashok Leyland), Pune-Chakan (Tata Motors, Mercedes, VW, Bajaj), Gurugram-Manesar (Maruti Suzuki, Hero MotoCorp), Sanand (Tata EV hub), Bangalore (Bosch, Toyota). India: 3rd largest automobile market globally.

IT Hubs

Bangalore (Silicon Valley — 35%+ of IT exports), Hyderabad (Cyberabad), Pune (Hinjewadi), Chennai (OMR), Noida/Gurugram (NCR), Mumbai (Navi Mumbai). Tier 2: Coimbatore, Kochi (Infopark), Ahmedabad (GIFT City), Jaipur, Indore, Bhubaneswar.

Key Pattern: Industrial regions follow resource availability (Jharkhand mineral belt), port access (Mumbai, Chennai, Gujarat), or policy/skill concentration (Bangalore IT). For GS1 answers, connect industrial location to these three factors — natural, locational, and institutional.

7. National Parks & Tiger Reserves

State-wise Key National Parks

State	Key National Parks	Famous For
Madhya Pradesh	Kanha, Bandhavgarh, Pench, Satpura, Panna	Highest tiger population (785). Kanha — Jungle Book inspiration. Bandhavgarh — highest tiger density.
Karnataka	Bandipur, Nagarhole, Bannerghatta, Kudremukh	2nd highest tiger population. Bandipur — part of Nilgiri Biosphere.
Uttarakhand	Jim Corbett (oldest, 1936), Rajaji, Nanda Devi, Valley of Flowers (UNESCO)	Corbett — first tiger reserve under Project Tiger (1973).
Assam	Kaziranga (UNESCO), Manas (UNESCO), Dibru-Saikhowa, Orang	Kaziranga — world's largest one-horned rhino population (70% of global). Manas — pygmy hog, wild buffalo.
Rajasthan	Ranthambore, Sariska, Keoladeo (UNESCO, birds), Desert NP	Ranthambore — famous tiger sightings. Keoladeo — Siberian crane migratory.
Kerala	Periyar, Silent Valley, Eravikulam	Silent Valley — biodiversity hotspot, never disturbed. Eravikulam — Nilgiri Tahr.
Gujarat	Gir (Asiatic lions), Blackbuck NP, Marine NP (Jamnagar)	Only wild Asiatic lion population globally (674 lions).
West Bengal	Sunderbans (UNESCO), Buxa, Gorumara, Singalila	Sunderbans — Royal Bengal tiger, world's largest mangrove forest.
Odisha	Similipal (UNESCO core), Bhitarkanika (mangroves), Chilika (lake, birds)	Similipal — melanistic tigers. Bhitarkanika — saltwater crocodiles. Chilika — largest brackish water lagoon.

Tiger Reserves — Top 10 by Population (2022)

1. **Bandipur** (Karnataka) — 150+
2. **Nagarhole** (Karnataka) — 140+
3. **Bandhavgarh** (MP) — 135+
4. **Kanha** (MP) — 120+
5. **Jim Corbett** (Uttarakhand) — 110+
6. **Kaziranga** (Assam) — 100+
7. **Ranthambore** (Rajasthan) — 80+
8. **Mudumalai** (TN) — 70+
9. **Pench** (MP, MH) — 70+
10. **Tadoba Andhari** (Maharashtra) — 65+

Map Tip for NP & TR: Mark these 6 on your India outline: (1) Jim Corbett (Uttarakhand, north) — oldest TR, (2) Gir (Gujarat, west) — only Asiatic lion, (3) Kaziranga (Assam, NE) — one-horned rhino, (4) Sunderbans (WB, east) — Royal Bengal tiger, (5) Ranthambore (Rajasthan) — tiger, (6) Periyar (Kerala, south) — elephant reserve. Covers all directions in one map.

8. Ports, Airports & SEZs

Major Ports of India (12 Major)

Port	State	Type	Significance
Kandla (Deendayal)	Gujarat	Tidal	Key crude oil import. Adani Mundra (largest private port, 35%+ container traffic).
Mumbai (JNPT)	Maharashtra	Natural, Deepest	Largest container port (JNPT handles 50%+ of India's containerised trade).
Chennai (Kamarajar)	Tamil Nadu	Artificial	2nd largest container. Chennai — oldest artificial port. Ennore — first corporatised major port.
Kolkata-Haldia	West Bengal	Riverine	Only riverine major port. Haldia — deep draft satellite.
Visakhapatnam	Andhra Pradesh	Deepest Landlocked	Iron ore, crude. Eastern Naval Command strategically based here.
New Mangalore	Karnataka	Major	Iron ore export, crude for Panambur refinery.
Cochin (Kochi)	Kerala	Natural	ICTT Vallarpadam — international container transshipment terminal. LNG terminal.
Paradip	Odisha	Major	Iron ore export (Bailadila).
VO Chidambaranar (Tuticorin)	Tamil Nadu	Artificial	Gateway for South TN. Coal, edible oil.

Key Airports

Airport	City	Passenger Traffic (approx)
Indira Gandhi International	Delhi	~75 million/year (busiest)
Chhatrapati Shivaji Maharaj	Mumbai	~55 million/year
Kempegowda International	Bengaluru	~40 million/year
Rajiv Gandhi International	Hyderabad	~25 million/year
M. R. Pamban International	Chennai	~25 million/year
Netaji Subhas Chandra Bose	Kolkata	~23 million/year

Special Economic Zones (SEZs)

SEZ Act 2005: 380+ notified SEZs. Key ones: SEEPZ (Mumbai — electronics, gems & jewellery), Noida SEZ (IT/ITeS), Mahindra World City (Chennai — auto, IT), Cochin SEZ (IT, electronics), Visakhapatnam SEZ, Surat SEZ (diamond, textiles), Jaipur SEZ. Top sectors: IT/ITeS (70%+ of exports), engineering, pharma. Exports: Rs 7+ lakh crore (2023-24).

9. International Borders & Strategic Locations

India's Land Borders

Neighbour	Border Length	Bordering States	Key Points
China	3,488 km (disputed LAC)	Ladakh, UK, HP, Sikkim, Arunachal (5)	Galwan (2020 clash), Doklam tri-junction, Pangong Tso, Tawang (claimed by China), Nathu La (border trade)
Pakistan	3,323 km	J&K, Ladakh (POK), Punjab, Rajasthan, Gujarat	Wagah (Attari — only land crossing), LOC (740 km), Sir Creek (maritime dispute)
Bangladesh	4,096 km (longest)	WB, Assam, Meghalaya, Tripura, Mizoram (5)	Petrapole (largest LCS), LBA 2015 (enclaves exchanged), fencing mostly complete
Nepal	1,751 km	UK, UP, Bihar, WB, Sikkim (5)	Raxaul (largest trade point), open border, Kalapani dispute, Pancheshwar project
Myanmar	1,643 km	Arunachal, Nagaland, Manipur, Mizoram (4)	Moreh (Manipur — largest trade), FMR up to 16 km (being reviewed), India-Myanmar-Thailand highway
Bhutan	699 km	Sikkim, WB, Assam, Arunachal (4)	Phuntsholing (gateway from WB), Doklam tri-junction with China

Critical Note: Most frequently tested border facts: (1) Longest border — Bangladesh, (2) China border — 5 states, (3) Pakistan border — 4 states, (4) Myanmar border — 4 states, (5) Nepal border — 5 states, (6) Bangladesh border — 5 states. Know the number of states each neighbour touches — a common Prelims and GS1 question.

Strategic Locations

- **Andaman & Nicobar Islands:** Indira Point (southernmost point of India), Campbell Bay, Great Nicobar. Strategic for India's Act East policy and Malacca Strait monitoring. New base INS Baaz.
- **Lakshadweep:** 36 islands, 12 atolls. Minicoy — southernmost atoll. Agatti — airport. Strategic for Arabian Sea SLOCs.
- **Kutch (Gujarat):** Rann of Kutch — border with Pakistan. Sir Creek dispute (maritime boundary). Koteswar — westernmost point of India.
- **Siachen Glacier:** World's highest battleground (5,400-7,300m). Operation Meghdoot (1984). Salto Ridge — strategic for Pakistan's Karakoram Highway overlook.
- **Doklam (India-China-Bhutan tri-junction):** 2017 standoff. Strategic location connecting India's Chicken's Neck corridor to NE states.
- **Chicken's Neck (Siliguri Corridor):** 22 km wide, connects NE India to mainland. Strategic vulnerability. Doklam and Sikkim importance relates to this.
- **Wakhan Corridor (Afghanistan):** 106 km border with Afghanistan via Wakhan. De facto defunct but strategically relevant.

International Border Facts for GS1: India shares land borders with 7 countries (China, Pakistan, Nepal, Bhutan, Bangladesh, Myanmar, Afghanistan). Maritime neighbours: Sri Lanka, Maldives, Indonesia, Thailand, Myanmar. Total coastline: 7,516.6 km (mainland 5,422.6 km + island territories 2,094 km). Longest coastline state: Gujarat (1,600 km).

10. Quick-Reference: Map Facts for GS1 Answers

10 Geographical Facts Worth Memorising

Fact	Details
Standard Meridian of India	82 deg 30' E (Mirzapur, UP) — IST reference line. Passes through 5 states: UP, MP, Chhattisgarh, Odisha, AP.
Tropic of Cancer	23 deg 30' N. Passes through 8 states: Gujarat, Rajasthan, MP, Chhattisgarh, Jharkhand, WB, Tripura, Mizoram.
Northernmost point	Indira Col (Siachen Glacier, Ladakh) — disputed with Pakistan
Southernmost point	Indira Point (Great Nicobar, A&N Islands) — submerged in 2004 tsunami, now just above water
Easternmost point	Kibithu (Anjaw, Arunachal Pradesh) — near China-Myanmar border
Westernmost point	Koteshwar (Kutch, Gujarat)
Highest peak in India	Kanchenjunga (8,586m, Sikkim) — 3rd highest globally
Largest delta	Sunderbans (Ganga-Brahmaputra delta) — world's largest (covers India and Bangladesh)
Largest river island	Majuli (Assam, Brahmaputra) — shrinking due to erosion
Largest saline water lake	Sambhar (Rajasthan) — also important for salt production

Key Dams Quick Reference

Dam	River	State	Distinction
Tehri	Bhagirathi	Uttarakhand	Highest dam in India (260.5m)

Bhakra Nangal	Sutlej	Himachal/ Punjab	2nd highest in Asia (226m); 'New Temple of Resurgent India' — Nehru
Hirakud	Mahanadi	Odisha	Longest dam in India (4.8 km)
Sardar Sarovar	Narmada	Gujarat	Largest concrete gravity dam; part of Narmada Valley Project
Nagarjunasagar	Krishna	Telangana	Largest masonry dam; 124m height
Kallanai (Grand Anicut)	Cauvery	Tamil Nadu	Oldest dam in India (2nd century AD — Karikala Chola)
Polavaram	Godavari	Andhra Pradesh	Multi-purpose; national project status; contentious with Odisha

Map Integration Strategy for GS1 Answers: Every geography answer must have a hand-drawn map or diagram. For a River System question: draw India outline + 3 Himalayan rivers + 3 peninsular rivers. For a Climate question: draw India + rainfall zones with colour. For Agriculture: show crop regions in the map itself. A 2-minute map effectively communicates 10 minutes of text.

The 'Big Three' Maps to Practise: (1) India Physical — rivers, mountains, passes, (2) India Agriculture — crop regions, (3) India Resources — minerals, oil, ports. Practise each 10 times with a pencil and scale. In the exam, reproduce the relevant elements from your mental template. These three maps cover 80%+ of GS1 geography map questions.

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Data compiled from Survey of India, Ministry of Mines, Ministry of Agriculture, Ministry of Environment, and Government geographical surveys.